For any matrix A, by rank nullity thm,

$$r(A) + Null(A) = \dim(\mathbb{R}^n) \quad ; \; A : \mathbb{R}^n \to \mathbb{R}^n.$$ — (I)

$$Im(A) \cap Null(A) = \{0\} \quad (\text{why?})$$ — (II)

take any $v \in Im(A) \cap Null(A)$

$A(v) = v.$

$\Rightarrow 0 = v$.

$v \in Im(A)$

$\Rightarrow \exists\, v' \in \mathbb{R}^n$ s.t.

$Av' = v.$

$\Rightarrow A^2 v' = Av$

$\Rightarrow Av' = Av = v.$

From (I) & (II)

$$\mathbb{R}^n = Im(A) \oplus Null(A)$$

Let's consider $\dim(Im(A)) = r$ & $\dim(Null(A)) = n - r.$

here $1 \leq r \leq n-1$ $[r \neq n \;\&\; r \neq 0]$

If $r = n$.

$A(A - I) = 0$ $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \neq A$

$0, 1$

$\Rightarrow \Leftarrow$

If $r = 0$. $\Rightarrow A = 0$ $\Rightarrow \Leftarrow$

given, $AB = BA$,

claim: $B(\text{Im}(A)) \subseteq \text{Im}(A)$ & $B(\ker(A)) \subseteq \ker(A)$

pf: take $v \in \text{Im}(A)$

$\Rightarrow \underbrace{B(v)} = B(A(v)) = A\underbrace{B(v)}$

$\Rightarrow B(v) \in \text{Im}(A)$

$v \in \ker(A)$

$\Rightarrow A(B(v)) = B(A(v)) = B(0) = 0$.

$\Rightarrow B(v) \in \ker(A)$.

$\mathbb{R}^n = \underline{\text{Im}(A)} \oplus \underline{\text{Null}(A)}$

$[B]_{\mathcal{B}} = \begin{pmatrix} B_1 & 0 \\ 0 & B_2 \end{pmatrix}$

$B^m = 0$ given.

$\boxed{B_1^m = 0 \ \& \ B_2^m = 0}$

$B^m = \begin{pmatrix} B_1^m & 0 \\ 0 & B_2^m \end{pmatrix}$

$= 0$

size$(B_1) = r \times r \Rightarrow B_1^r = 0$

size$(B_2) = (n-r) \times (n-r) \Rightarrow B_2^{n-r} = 0$.

nilpotency of $B = \max\{r, n-r\}$

$\leq \underline{n-1}$ $[0 \leq r \leq n-1]$

$\Rightarrow B^{n-1} = 0$